

Content-Preserving, Cognitive-Load-Aware Generation of Educational Slide Decks

Learnova research track — umbrella paper. Author list TBD.

Companion documents: PSF_DESIGN.md (PSF/CLASS in full), VISUAL_INTELLIGENCE.md (the decision layer), paper_draft.md (the PSF/CLASS-only submission).

Abstract

Systems that turn a document, syllabus, or set of notes into a slide deck have converged on one of two designs: **template filling**, which is rigid and ignores the shape of the content, or **large-language-model (LLM) generation**, which silently paraphrases, compresses, drops list items, and occasionally invents facts, with no way for a user to verify what survived. Neither design has a principled notion of whether a generated slide is *good for learning*, and both paginate content with fixed caps that truncate material.

We present **Learnova**, an open-source engine built around three ideas. **(1) A deterministic decision layer.** Every structural choice — which sentences are reproduced verbatim, which visual modality a slide takes, whether a bitmap figure is kept or redrawn, how many ideas go on a slide — is made by an explicit, inspectable rule with a stated guarantee, not by a learned model and not by the LLM. The LLM is used only for *rewording within constraints*, and its output is reconciled against a deterministic extractive lower bound, giving a **verifiable content-preservation property**: no source enumeration item is lost and no visual data is fabricated. **(2) Pedagogical Slide Fitness (PSF)**, a slide-quality metric whose form is derived from Cognitive Load Theory and Mayer's Cognitive Theory of Multimedia Learning and whose low-dimensional weight vector is calibrated against human ratings. **(3) CLASS**, a dynamic program that paginates a section across slides to minimise total cognitive-load cost — using PSF's load term as its cost function — generalising Knuth & Plass optimal line breaking and provably dropping no content.

The deterministic layer runs in milliseconds with no inference cost; PSF and CLASS add none. We describe the design, state the guarantees, and give a pre-registered evaluation plan on open educational material: content-fidelity against LLM baselines, PSF's agreement with human quality judgements, VMS's agreement with human "right visual?" judgements, and blind deck preference for CLASS over even splitting.

1. Introduction

Automatic "text-to-deck" generation is now a mainstream feature of office suites and a popular use of general-purpose LLMs ("make me a presentation about X"). Three problems remain unsolved.

The fidelity gap. An LLM asked to "make slides from this chapter" performs *abstractive* transformation: it rewrites, and in rewriting it under-generates. A passage that lists "the five stages of X" routinely emerges with two or three; a two-word list item ("Machine Translation") is dropped as

too short to be a bullet; a definition is loosely paraphrased; occasionally a figure caption or a statistic is confabulated. Summarisation research names these *content-selection* and *faithfulness* errors [Maynez et al. 2020; Kryściński et al. 2020], and for a *study aid* they are disqualifying: the student cannot tell that the deck is lossy, and the missing item is exactly the one the exam asks about. Commercial deck tools do not report what they changed.

The evaluation gap. Systems optimise for "looks like slides", not for learning. Where an automatic quality score exists it is an *asserted* additive sum over surface features — text density, bullet count, presence of a takeaway — with hand-picked weights and thresholds, no theoretical model behind the choice of terms, and no validation against any human judgement of learning quality. Such a score rewards superficial regularity: a beautiful empty slide can score well.

The segmentation gap. How much content goes on each slide is decided by a fixed per-profile cap plus an even rebalance. A cap silently truncates. Even splitting ignores that a slide's difficulty is non-linear in its length and depends on how many of its ideas *interact* — the central claim of Cognitive Load Theory.

Our position. The three gaps are one design problem. If a system can *measure* the cognitive load a slide imposes, it can also *paginate* to minimise it — the metric and the planner should be the same model. And the structural decisions that determine fidelity (what to keep, what to draw, what to split) should be made by rules that carry guarantees, with the LLM confined to rewording, so that "nothing was lost" is a property of the system rather than a hope.

Contributions.

1. **A deterministic decision layer** (§4) for document→deck transformation: register-protected verbatim text policy, an explainable Visual Modality Selector, an image-action policy, enumeration-atomic content preservation, a never-fabricate downgrade for visuals, and an extractive reconciliation step that lower-bounds the LLM. We state and prove a content-preservation property (Guarantee 1) and a no-fabrication property (Guarantee 2).
2. **PSF** (§5): a theory-derived, human-calibrated slide-fitness metric with a multiplicative (Cobb–Douglas) form, replacing the asserted engagement heuristic.
3. **CLASS** (§6): an optimal, content-preserving slide-segmentation dynamic program whose objective is PSF's load term — a Knuth–Plass generalisation.
4. **System** (§7): a web-first render-and-capture pipeline, a shared Gallery of pre-built decks, and a grounded conversational assistant, all consuming the same typed deck representation.
5. **A pre-registered evaluation plan** (§8) on open educational documents.

Learnova is open source; the decision layer, PSF, and CLASS are pure Python with no network dependency and are unit-tested, including a brute-force optimality check for CLASS.

2. Background and related work

Cognitive Load Theory (CLT) [Sweller 1988; Sweller, Ayres & Kalyuga 2011]. Working memory holds ~4 elements [Cowan 2001]. CLT decomposes the load a learning artefact imposes into *intrinsic* (element interactivity — how many items must be held simultaneously to understand the material;

managed by sequencing and **segmenting**), *extraneous* (presentation choices that add load without learning — a wall of text, split attention, redundancy; reduced by layout), and *germane* (effort that builds schemas). A good slide minimises extraneous load, manages intrinsic load by segmenting, and leaves working-memory headroom for germane processing.

Cognitive Theory of Multimedia Learning (CTML) [Mayer 2009]. Empirically grounded principles, several of which are *measurable from slide content*: **coherence** (exclude material that does not serve the goal), **signalling** (cue the structure — headings, a stated takeaway), **redundancy** (do not reproduce on-screen text verbatim in narration; do not repeat), **spatial contiguity** (put a figure next to its referring text), **segmenting** (deliver in learner-paced segments).

Automatic presentation generation. Doc2ppt-style and template systems [e.g. Fu et al. 2022; Sefid et al. 2021] and, more recently, LLM deck tools. To our knowledge none reports a validated quality metric, an optimal segmentation, or a content-preservation guarantee.

Faithfulness in summarisation [Maynez et al. 2020; Kryściński et al. 2020; Pagnoni et al. 2021]. Establishes that abstractive models drop and hallucinate content and that the practical mitigation is to treat certain spans as atomic and to check the output against an extractive reference. We apply both to list-bearing spans and to whole slides.

Optimal segmentation / line breaking [Knuth & Plass 1981]. A one-dimensional dynamic program over prefix break points is globally optimal when the cost of a segment depends only on its contents. We reuse this structure with a cognitive-load cost (§6).

Information-theoretic content measures. idf-weighted content and surprisal as proxies for "how much a span says" [e.g. Peyrard 2019]. PSF uses a diminishing-returns function of the summed rarity of *newly introduced* concepts as its upside term (§5.2).

Readability metrics (Flesch–Kincaid and relatives) are text-only and say nothing about layout, visuals, or load interaction; PSF subsumes a readability signal as one driver among several.

3. System overview

Learnova is a staged pipeline from a source document to two artefacts (an animated PowerPoint file and a self-contained interactive web deck). The stages are: **convert** (PDF/PPTX/typed text → a Markdown intermediate representation, with figures extracted and re-anchored to the section that discusses them); **chunk** (split on heading boundaries, merge fragments, explode enumerations — §4.2); **improve** (per-chunk: LLM reword *or* deterministic extractive structuring, then extractive reconciliation — §4.6); **visual plan** (VMS — §4.4 — assigns a treatment and its structured data); **image policy** (§4.5); **density** (CLASS — §6 — paginates overflow); **quiz** (checkpoint questions interleaved at run boundaries); **score** (PSF — §5); **build** (render both artefacts).

Two properties hold across the pipeline by construction:

- The system runs end-to-end **with no LLM key at all** (every LLM stage has a deterministic fallback), so all guarantees below are properties of the deterministic path and are only *tightened*, never relaxed, by the LLM.

- The deck is carried as a single **typed representation** consumed identically by the PPTX builder, the web builder, the Gallery, and the assistant.

4. The deterministic decision layer

*This is the component that is **not** a statistical model and **not** a closed-form mathematical function: it is a set of explicit decision procedures, each with a stated guarantee. The LLM is a constrained sub-routine, not the decision-maker.*

4.1 Design principle

For each structural question the pipeline asks — *keep this sentence as written? draw this as a diagram? which diagram? keep this bitmap? how many ideas on this slide?* — Learnova commits to three properties:

1. **Explainability.** The decision is a sum of named feature contributions or a short rule chain; the Studio UI can display *why* a slide became a pie chart.
2. **A conservative default.** When no rule fires with enough evidence, the system chooses the option that cannot lose information: keep the text, keep the image, do not split.
3. **A guarantee.** Each sub-policy has a property that holds regardless of the LLM's behaviour (stated below).

The LLM's role is narrowed to *rewording within constraints* — turning a retrieved passage into slide-shaped prose without changing which facts, list items, or defined terms appear.

4.2 Enumeration-atomic content preservation

Module: ai/enumeration.py. The single largest fidelity failure we observed was content selection on lists. `extract_enumerations(text)` deterministically pulls every enumeration in a passage:

```
Enumeration(lead="The five phases of NLP", items=[...verbatim...],
            kind="phases", claimed_count=5, style="numbered")
```

It recognises numbered runs, bulleted runs, inline series ("X, Y, and Z"), header-style item lists, and plain paragraph blocks introduced by a plural head noun ("phases | stages | methods | applications | advantages | ..."). Items are captured **verbatim**, de-duplicated, order preserved; the claimed count ("five") is parsed and checked against the item count.

Downstream:

- `split_into_item_sections` turns a *definitional* enumeration (each item a short heading followed by its own explanation) into an overview slide plus one slide per item ("Stage 2 of 5: Syntactic Analysis"), which teaches better than a wall of bullets and animates naturally one item per click.
- `missing_items(enum, covered_text)` reports which items the improved slide fails to mention; the improver force-appends them.
- Neither scoring, nor length filters, nor the LLM is permitted to alter an enumeration's *membership*.

Guarantee 1 (enumeration preservation). *For every enumeration e detected in a source section, every item of e appears in the generated deck, verbatim or as a normalisation-equivalent string.* The item set is copied before any lossy stage and re-checked after; a missing item is re-inserted. Empirically this raised item retention on a graduate NLP chapter from 1–2 of 5 to 5 of 5 across every enumeration in the document.

4.3 Register-protected verbatim text policy

Module: `ai/text_policy.py`. `classify_sentences(text)` labels each sentence **VERBATIM**, **MERGE**, or **TIGHTEN**:

- **VERBATIM** — definitional register (is defined as, X is the measure of, denoted by), theorem / law statements, anything in quotation marks or following wrote:, legal register (shall, pursuant to, Section 3), code (`{}`, `=>`, `def`, `SELECT ... FROM`), and formula-in-prose ($E = mc^2$, α , $\leq$).
- **MERGE** — a near-duplicate (string containment or Jaccard ≥ 0.72) of an earlier sentence.
- **TIGHTEN** — everything else, and only this class is exposed to the trimming / rewording machinery.

`protect_verbatim(text)` marks the VERBATIM spans so the pipeline's length-based trimming and the LLM prompt both leave them untouched. Precise wording — the part a student must reproduce on an exam — is never paraphrased.

4.4 Visual Modality Selection (VMS)

Module: `ai/visual_selector.py`. VMS is the deterministic answer to "which visual, when". It scores each of 20 treatments —

KEEP_TEXT · BULLETS · DEFINITION · QUOTE · FLOWCHART · CYCLE · TIMELINE · COMPARISON_TABLE · PROS_CONS · MATRIX_2X2 · PYRAMID · VENN · MIND_MAP · CARD_GRID · BAR_CHART · LINE_CHART · PIE_CHART · METRIC · IMAGE_FOCUS · MINIMAL_TEXT

— from *structural features of the content*, not keywords:

Treatment	Fires on
FLOWCHART	≥ 3 ordered steps; bonus for decision cues ("if... then", "otherwise")
CYCLE	ordered steps and loop cues ("repeat", "feedback", "the cycle continues")
TIMELINE	≥ 2 real dates/years — not ordinal words alone
COMPARISON_TABLE	an extracted A-vs-B comparison, or ≥ 1 "vs / whereas / in contrast"
PROS_CONS	≥ 2 advantage/disadvantage cues for one subject
PIE_CHART	≥ 3 percentages summing to ≈ 100
LINE_CHART	a numeric series with time cues

Treatment	Fires on
BAR_CHART	≥ 3 numbers across categories (suppressed if a valid pie exists)
METRIC	exactly one figure and < 45 words
PYRAMID / VENN / MATRIX_2X2	level cues / shared-set cues / two-axis cues
MIND_MAP	≥ 4 loosely related concepts and no other structure

Every treatment carries a rationale string and the full per-feature scores map. The winning score must exceed a threshold $\tau_{\text{vms}} = 2.5$; **otherwise VMS returns KEEP_TEXT / MINIMAL_TEXT**.

Design property (defer-to-text). *A forced diagram is worse than clean text.* VMS emits a non-text treatment only when the content's structure independently clears τ_{vms} ; the null hypothesis is "this is prose".

VMS also serves as a **validation layer** over an LLM that is offered the same taxonomy: if the LLM proposes FLOWCHART but VMS scores no ordered steps, the proposal is rejected and the deterministic choice stands.

4.5 Image-action policy

Module: ai/image_policy.py. For each extracted figure, `decide_image_action(ImageMeta)` chooses from KEEP_AS_IS, CAPTION_ONLY, ENHANCE, REGENERATE, SUMMARISE_TO_STRUCTURE, or DROP using pixel size, aspect ratio, OCR word density, structure markers in the OCR ($\rightarrow$, step N, | ... | ..., yes/no), whether the figure is referenced in the text, and its topical relevance to the surrounding slide. The consequential case is **SUMMARISE_TO_STRUCTURE**: a bitmap of a flowchart is re-recognised (its OCR is run back through VMS) and rendered as a *native* Learnova flowchart, and the blurry screenshot is dropped.

4.6 Extractive reconciliation — a deterministic lower bound on the LLM

Module: ai/improver.py. Each chunk is structured two ways: by the LLM (`classify_and_structure_chunk`) and by a deterministic extractive summariser (`ai/extractive.py`, which performs no length-based sentence cutting and no per-bullet compression). Let w_{llm} and w_{ext} be the content-word counts of the two results. If

$$w_{\text{llm}} < 0.95 \cdot w_{\text{ext}}$$

the LLM under-delivered and its bullets are replaced (or back-filled) from the extractive result. The threshold 0.95 is a configurable retention floor (`LEARNOVA_MIN_RETENTION`). This makes the extractive path a *floor*: the deck is never more lossy than deterministic extraction, and is usually better-worded than it.

Guarantee 2 (no fabricated visual data). *A COMPARISON_TABLE with no parseable rows and a METRIC with no readable quantity are downgraded to plain text rather than rendered with invented cells or the literal words "Key Stat" at headline size.* The renderer (`rendering/deck_director.py`)

refuses to draw a lossy diagram family over a content-heavy slide or one whose bullets are full sentences (>12 words); it substitutes MINIMAL_TEXT.

4.7 Progressive-reveal grammar

Function: plan_reveal_groups. The build order of a slide's elements is a deterministic function of its treatment: one idea per step for lists / flowcharts / timelines; row-by-row for comparison tables; two halves for pros-cons and Venn; a single step for QUOTE / METRIC / DEFINITION; the takeaway is always last; groups are **consecutive and never reordered**. The web deck realises groups as Reveal.js fragments; the PPTX builder as per-shape entrance animations. Gallery decks ship with reveal enabled by default so a downloaded deck can be taught one point at a time.

5. Pedagogical Slide Fitness (PSF)

Module: scoring/psf.py. Full derivation: PSF_DESIGN.md.

5.1 Per-slide model

$$\text{PSF}(s) = E(s)^{\alpha} \cdot (1 - L(s))^{\beta} \cdot C(s)^{\gamma}, \quad \alpha + \beta + \gamma = 1$$

A **multiplicative (Cobb–Douglas)** form is deliberate: a slide that collapses on *any one* dimension — says nothing, is unreadable, or is incoherent — should score near zero regardless of the other two. An additive model lets a beautiful empty slide score 66/100, which is the failure mode of the heuristic PSF replaces.

E(s) — information efficiency. With a corpus-free rarity estimate $\hat{\text{idf}}(c) \in [0.4, 3.0]$ (from stop-word membership, a high-frequency-word set, and token length — technical terms are systematically longer):

$$\begin{aligned} U(s) &= \sum_{\{c \in \text{newconcepts}(s)\}} \hat{\text{idf}}(c) && \text{(delivered novel information)} \\ R(s) &= \text{mean_bullets} \max(\text{Jaccard}(\text{bullet}, \text{title}), \text{Jaccard}(\text{bullet}, \text{other bullets})) \\ E(s) &= (U(s) / (U(s) + \kappa_E)) \cdot (1 - R(s)) \end{aligned}$$

$U / (U + \kappa_E)$ is a Michaelis–Menten saturation: the first few rare concepts earn most of the score (a slide has *one* job). $\text{newconcepts}(s)$ excludes concepts introduced on an earlier slide, so restating does not score.

L(s) — cognitive load $\in [0, 1]$. Four drivers, each soft-clipped by a reference constant:

driver	definition	reference
element interactivity $\hat{e}$	$n_{\text{concepts}} \cdot (1 + \text{relational_cue_density}) / e_{\text{ref}}$	CLT element interactivity
extraneous text load $\hat{t}$	$\text{words} / t_{\text{ref}}$	working-memory chunk budget
visual complexity $\hat{v}$	$\max(\text{table_rows}, \text{flow_nodes}, \text{grid_cards}) / v_{\text{ref}}$	—
split attention $\hat{s}$	1 if a figure is described in text	contiguity principle

driver	definition	reference
	but placed on another slide	

$$L(s) = \sigma(w_e \cdot \hat{e} + w_t \cdot \hat{t} + w_v \cdot \hat{v} + w_s \cdot \hat{s} - \theta_L)$$

relational_cue_density counts connectives ("because", "therefore", "whereas", "leads to") — a text proxy for how many elements *interact*, which CLT says drives intrinsic load, not raw count.

C(s) — multimedia coherence $\in [0,1]$. Mean of four CTML indicators: signalling (a real title *and* an explicit takeaway); coherence (fraction of sentences whose overlap with the slide's concept centroid exceeds τ — a "seductive detail" detector); non-redundancy (figure caption $\neq$ bullet text); spatial contiguity (figure on the same slide as its referring text).

5.2 Deck model

$$\text{PSF}_{\text{deck}} = (\prod_s \text{PSF}(s))^{\{1/N\}} \cdot \text{Flow}(\text{deck})$$

Flow is a trapezoidal reward for *moderate* concept overlap between consecutive slides (Jaccard in $[\phi_{\text{lo}}, \phi_{\text{hi}}] \approx [0.18, 0.55]$): near 0 is a jarring topic jump; near 1 is repetition.

5.3 Calibration

Only the low-dimensional vector $(\alpha, \beta, \gamma, w_e, w_t, w_v, w_s, \theta_L)$ is learned; the scale constants $(\kappa_E, e_{\text{ref}}, t_{\text{ref}}, v_{\text{ref}})$ are fixed from theory / corpus statistics to keep the model identifiable. We fit by constrained non-linear least squares against mean human ratings, with (α, β, γ) Dirichlet-reparameterised onto the simplex, under 5-fold cross-validation. `scripts/calibrate_psf.py` performs the fit; the module ships literature-derived priors (Appendix A) so it is usable before any data is collected.

The paper's claim about PSF: the *shape* of the model is theory; only a weight vector is learned, and it is learned against human judgement with cross-validation and a held-out test set — in contrast to the asserted heuristic it replaces.

6. CLASS — Cognitive-Load-Aware Slide Segmentation

Function: `scoring/psf.py::segment_blocks`.

6.1 Problem

A section yields an ordered list of atomic content blocks $b_1 \dots b_m$ (bullets; for tables and flowcharts, rows and steps). Choose cut points $0 = i_0 < i_1 < \dots < i_k = m$ to

$$\begin{array}{ll} \text{minimise} & \sum_{j=1..k} \text{cost}(b_{\{i_{j-1}\}+1} \dots b_{i_j}) + \lambda \cdot k \\ \text{subject to} & \text{group size} \leq P_{\text{max}} \quad (\text{layout-safety hard cap}) \end{array}$$

with the per-slide cost taken **directly from PSF's load model**:

$$\text{cost}(g) = L(\text{slide}(g)) + \mu \cdot (1 - E(\text{slide}(g))) + v \cdot \text{overflow}(g)^2$$

$\lambda \cdot k$ regularises slide count (ties break toward fewer slides); $\mu \cdot (1-E)$ stops the optimiser emitting near-empty slides; $v \cdot \text{overflow}^2$ is a soft quadratic penalty past a target size *below* the hard cap.

This is structurally identical to Knuth–Plass optimal line breaking: "line" $\rightarrow$ "slide", "line badness" $\rightarrow$ "slide cognitive load", "hyphenation penalty" $\rightarrow \lambda$.

6.2 Algorithm and guarantees

```
CLASS(blocks b[1..m], P_max,  $\lambda$ ):
    best[0] = 0 ; cut[0] = 0
    for i in 1..m:
        best[i] = + $\infty$ 
        for p in max(0, i - P_max) .. i-1:
            c = best[p] + cost(b[p+1..i]) +  $\lambda$ 
            if c < best[i]: best[i] = c ; cut[i] = p
    reconstruct groups by following cut[] back from m
```

$O(m \cdot P_{\max})$ time, $O(m)$ space; $P_{\max} \leq 8$, so linear in section length and negligible beside the LLM stages.

- **Completeness (content preservation).** The cut points partition $1..m$; every block lands on exactly one slide. bullets[:4]-style truncation is structurally impossible.
- **Optimality.** Because cost depends only on a contiguous block, the standard exchange argument [Knuth & Plass 1981 §3] gives a global minimum of the objective over all legal segmentations. tests/test_psf.py includes a brute-force check against exhaustive enumeration for small m .
- **No orphan continuation slides.** Including $\mu \cdot (1-E)$ in the cost removes the one-line-continuation failure mode that even-split code patches with an ad-hoc rebalance.

6.3 Integration

CLASS is a drop-in replacement for the even-split call inside the density stage (LEARNNOVA_USE_CLASS=1); the atomic-layout, table-header-repeat, and "figure stays on part 1" rules around it are unchanged.

6.4 The metric-as-cost-function design

PSF's load term L (and its emptiness term $1-E$) is *both* how a finished slide is judged *and* the cost CLASS minimises while building it. One model gives the generator a conscience and a planner.

7. System contributions

7.1 Web-first render and capture

Both artefacts are produced from the same typed deck. The web deck is a self-contained single HTML file (Reveal.js inlined, offline-safe) with content-adaptive auto-fit (measures rendered height against a reference stage and scales dense slides down / sparse slides up), entrance animations, a bullet-hierarchy renderer, and a two-column text+figure layout. Progressive reveal is a URL/embed toggle, off by default for reading and on for presenting.

7.2 The Gallery

A shared catalogue of ~1,000 teaching topics across ~30 subjects (GALLERY.md). Curated topics ship a structured teaching brief and a **pre-built deck**; a user opens it, adapts it, or downloads it without generating anything. Gallery decks are produced by a batch runner over the same pipeline with providers disabled, so they are fully deterministic and reproducible from the catalogue. This turns the fidelity and load machinery into a directly reusable artefact, and gives an evaluation corpus that is regenerable byte-for-byte.

7.3 Grounded conversational assistant

A chat + voice assistant (ASSISTANT_ARCHITECTURE.md) resolves natural language to typed actions over a deterministic 58-intent taxonomy with an LLM fallback for low-confidence utterances. Every action is validated server-side against the real deck library before it runs; an LLM-supplied identifier is resolved and ownership-checked, never trusted. Explanations are **grounded** in the retrieved deck/slide text — the reply may be simplified or translated, but the stored deck is never modified. The assistant is Gallery-aware: "do you have slides on X?" runs a catalogue search and, when a pre-built deck exists, confirms and opens it.

8. Evaluation plan (pre-registered)

8.1 Corpus and ratings

- **Documents.** Open educational material stratified by discipline: OpenStax chapters, MIT OpenCourseWare notes, arXiv survey introductions. Target 120–300 slides across 20–40 decks. The Gallery's 39 curated decks are a regenerable subset.
- **Slide ratings.** ≥ 3 raters per slide on a learning-oriented rubric (manageable load? clear single point? would it help a student?). Report inter-rater reliability (Krippendorff's α). This is the ground truth for PSF calibration and testing.
- **VMS ratings.** For a sample of slides, raters judge "is this the right visual for this content?" for the VMS choice, an LLM-only choice, and the legacy 5-type router.
- **Fidelity annotation.** For a sample of source sections, annotators mark every enumeration item and every defined term; we measure recall in the generated deck.

8.2 Studies

#	Question	Method	Metric
E1	Does PSF predict human slide-quality ratings better than the baseline heuristic and than ablations?	5-fold CV, held-out test	Spearman ρ , RMSE, NDCG@k
E2	Which PSF terms matter?	leave-one-out on E, L, C and on each load driver	$\Delta\rho$

#	Question	Method	Metric
E3	Multiplicative vs additive aggregation?	fit both, compare on test	Δ RMSE, calibration curves
E4	Does CLASS beat even-split?	within-subjects, same content, blind deck preference + PSF_deck + slide count	preference %, effect sizes
E5	Content fidelity vs LLM baselines.	enumeration-item recall and defined-term recall: Learnova (deterministic path), Learnova (LLM path + reconciliation), GPT-class "make me slides", a template tool	recall, hallucinated-fact rate
E6	VMS agreement with human "right visual?" vs LLM-only and legacy router.	rater agreement	accuracy, Cohen's κ
E7	(stretch) Learning outcome.	between-subjects quiz after a CLASS deck vs an even-split deck on the same content	mean quiz score, time-on-task
E8	Cost.	wall-clock of the decision layer, PSF, and CLASS across section lengths	ms

8.3 Hypotheses

- H1: PSF ρ with human ratings $>$ baseline heuristic ρ , on held-out test.
- H4: CLASS decks preferred $> 50\%$ and use $\leq$ even-split slide count.
- H5: Learnova enumeration-item recall ≈ 1.0 and $>$ LLM-baseline recall; hallucinated-fact rate ≈ 0 .
- H6: VMS agreement $\geq$ LLM-only agreement, both $>$ legacy router.

9. Threats to validity

- **Perceived vs actual learning.** Raters judge perceived quality; E7 addresses this partially with an outcome measure.
- **Noisy concept extraction.** Rarity and relational density come from a keyword extractor; we report robustness with true corpus idf and an embedding-based extractor as a second condition.

- **Rater pool and rubric.** Learning-quality judgements are subjective; we report α and use a pre-registered rubric.
 - **Domain generalisation.** We stratify the corpus by discipline and report per-stratum ρ .
 - **VMS coverage.** The 20-treatment taxonomy is not exhaustive (no maps, no music notation); out-of-taxonomy content correctly falls to text, which is safe but not always ideal.
 - **Guarantee scope.** Guarantee 1 covers *detected* enumerations; an enumeration written in an unusual style may be missed by the detector (not mis-handled — simply treated as prose). We report detector recall on the fidelity-annotated sample.
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10. Reproducibility

- **Decision layer:** `learnova.ai.{enumeration, text_policy, visual_selector, image_policy}` and `learnova.ai.improver` — deterministic, `stdlib` + `regex`, unit-tested (`tests/test_content_fidelity.py`, `tests/test_visual_intelligence.py`, `tests/test_visual_planning.py`).
 - **PSF / CLASS:** `learnova.scoring.psf` — deterministic, `stdlib` only; `tests/test_psf.py` includes a brute-force optimality check.
 - **Calibration:** `scripts/calibrate_psf.py ratings.csv --out params.json`.
 - **Deck scoring:** `score_all_slides(deck, engine="psf")`.
 - **CLASS in the pipeline:** `LEARNOVA_USE_CLASS=1`.
 - **Corpus:** the Gallery catalogue (`data/gallery/catalog.json`) plus the batch runner (`python -m learnova.gallery.builder --all`) regenerates the curated decks byte-for-byte with no network.
 - **No-LLM mode:** `LEARNOVA_NO_LLM=1` forces the deterministic path end-to-end; every claimed guarantee holds in this mode.
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11. Discussion and limitations

- **One model, two jobs.** PSF judging and CLASS building from the same load term is the paper's spine; it also means a calibration error propagates to both. E2/E3 quantify sensitivity.
 - **Where the decision layer is weak.** Sarcasm, discipline-specific notation, and figures we can only see through OCR/caption. The conservative default keeps these *safe* (text is kept) but not always *optimal*.
 - **The LLM still matters for prose quality.** Guarantees bound *what* appears, not how well it reads; E5 measures fidelity, not fluency, and the reconciliation step can make a deck faithful but flat.
 - **Extending CLASS.** Two-dimensional layout optimisation, cross-section reordering, learner-adaptive `P_max`.
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12. Conclusion

Turning a small amount of learning theory into an objective, and making the structural decisions with rules that carry guarantees rather than with a model that might drop things, gives an automatic slide generator a conscience, a planner, and a fidelity property a user can rely on. The decision layer, PSF, and CLASS are open source in Learnova, with a pre-registered evaluation on open educational material.

Appendix A — default parameters

scoring/psf.py: PSFParams: $\alpha=0.40$, $\beta=0.42$, $\gamma=0.18$; $w_e=1.10$, $w_t=1.35$, $w_v=0.85$, $w_s=1.60$, $\theta_L=2.30$; $\kappa_E=6.0$, $e_{ref}=9.0$, $t_{ref}=16.0$, $v_{ref}=8.0$; seductive $\tau=0.08$; flow band $[0.18, 0.55]$; $\lambda=0.35$, $\mu=0.9$, $\nu=0.25$. VMS threshold $\tau_{vms} = 2.5$. Retention floor $LEARNOVA_MIN_RETENTION = 0.95$.

Appendix B — notation

Symbol	Meaning
s, N	a slide; slide count
E, L, C	PSF sub-indices: information efficiency, cognitive load, coherence
α, β, γ	Cobb–Douglas exponents (simplex)
$U(s), R(s)$	delivered novel information; redundancy
$\hat{idf}(c)$	corpus-free rarity estimate of concept c
$\hat{e}, \hat{t}, \hat{v}, \hat{s}$	normalised load drivers
$b_1 \dots b_m$	atomic content blocks of a section
$P_{max}, \lambda, \mu, \nu$	CLASS hard cap; slide-count / emptiness / overflow weights
τ_{vms}	VMS minimum winning score to leave plain text

References (to be completed)

Cowan 2001 · Fu et al. 2022 · Knuth & Plass 1981 · Kryściński et al. 2020 · Mayer 2009 · Maynez et al. 2020 · Pagnoni et al. 2021 · Peyrard 2019 · Sefid et al. 2021 · Sweller 1988 · Sweller, Ayres & Kalyuga 2011.